

BOTTLENECK

Multi-Agent Supply-Chain Disruption & Recovery

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Objectives & Deliverables



OBJECTIVES
- Detect supplier failures from email and structured events
- Coordinate grounded recovery decisions across agents

DELIVERABLES
- Live Cloud Run dashboard with approval workflow
- Auditable ADK + CrewAI outputs, evaluations and evidence





Literature Review

PRODUCT / SERVICE | STRENGTH | GAP
SAP IBP | enterprise planning | limited agent transparency
Kinaxis | concurrent planning | proprietary orchestration
o9 | digital-brain analytics | costly deployment
Bottleneck | live agent evidence | academic synthetic scope
Sources: vendor documentation and framework specifications; full citations on Slide 11.



Multi-Agent Architecture

EMAIL / EVENT → ADK TRIAGE
↓ shared Session State
SUPPLIER — LOGISTICS — INVENTORY — PRODUCTION
parallel fan-out / join

ADK state supervisor → deterministic risk → conditional A2A supplier research → recovery planning → reflection → human approval
Patterns: sequential, parallel, routing, reflection, bounded replanning, memory, HITL.



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Agent–Tool Communicati on · MCP

ADK Supplier Agent —MCP stdio→ lookup_supplier / lookup_supplier_orders
ADK Logistics Agent —MCP stdio→ lookup_order_shipments / calculate_delay_days

Validated tool result → callback → evidence store → shared state
All tools are read-only, typed and source-labelled.

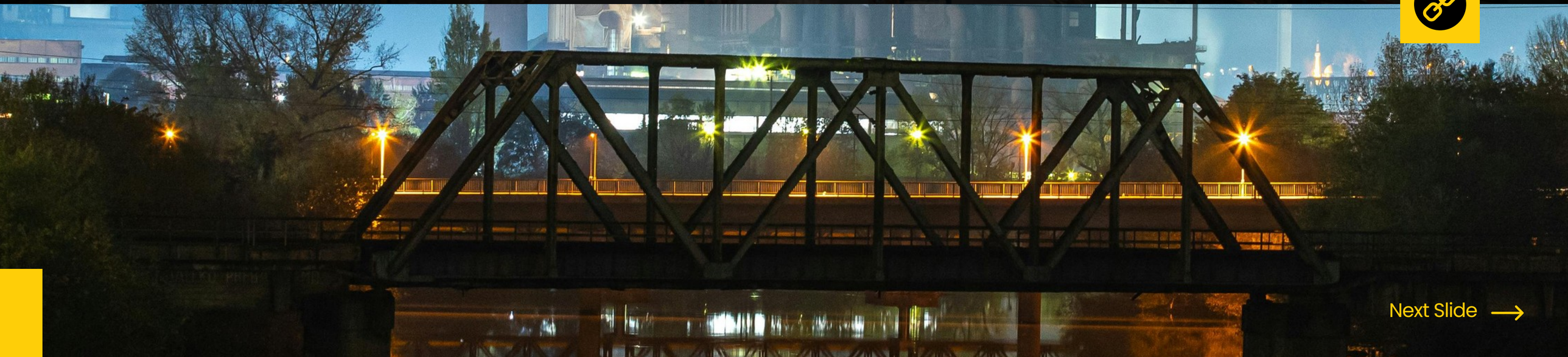
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Agent–Agent Communicati on · A2A

GOOGLE ADK Alternative-Supplier Agent
| JSON-RPC 2.0 · A2A v0.3

CREWAI A2A ENDPOINT /a2a
Researcher → Analyst → Reviewer

Agent Card: /a2a/.well-known/agent-card.json
Request: component + quantity + shared evidence
Response: Pydantic candidate list + role outputs
Interoperability boundary is separately addressable and auditable.



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Securing the Agentic System

INPUT GUARDRAILS
Prompt-injection patterns · threat/toxicity checks · input-size limits · secret redaction

TOOL SAFEGUARDS
Allow-listed read-only tools · typed arguments · schema validation · timeout/retry bounds

CALLBACKS & VALIDATION
before_model_callback · before_tool_callback · after_tool_callback
Pydantic contracts · constraint engine · reflection gate · human approval
Secrets remain in Google Secret Manager; no credential is committed.



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Deployment Architecture



USER BROWSER
| HTTPS
▼
GOOGLE CLOUD RUN · asia-south2
FastAPI + vanilla HTML/CSS/JS + ADK agents + CrewAI A2A + MCP server
├ Gemini API (Secret Manager)
└ revision-local seeded SQLite

CI/CD: GitHub main → Cloud Build trigger → Cloud Run revision
Zero-cost controls: min instances 0 · max instances 1 · request billing
Demo URL: bottleneck-545600829568.asia-south2.run.app

Team Responsibility / Self-Contribution

EQUAL CONTRIBUTION — 25% EACH
Aastha Thakur (230552) | Agent design, multi-agent architecture, orchestration, state management
Shivani (Enrollment) | MCP design, A2A communication, guardrails and callbacks
Mayank (Enrollment) | Observability, integration and testing, evaluation and documentation
Lokesh (Enrollment) | Interface UI, deployment, live-demo validation and release quality

All four members: equal 25% contribution. Insert the three pending enrollment numbers before submission.

Working System · Live Evidence

LIVE CLOUD RUN CASE
Email → incident b49a4b5e... → 28 agent/tool events → WAITING_FOR_APPROVAL
10-day delay · 1,280-unit shortage · 12.8-day modeled downtime
Selected recovery: SUP-B 450 + SUP-D 250 + SUP-C 300

EVALUATION
3/3 scenarios · 6/6 tools · 41/41 structured outputs
37.335 s end-to-end · $0.001609 estimated model cost





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Conclusions

- End-to-end supplier-failure recovery is deployed and independently demonstrable from Cloud Run.
- Google ADK shared-state orchestration and CrewAI specialist research interoperate through A2A.
- MCP tools, Pydantic contracts, callbacks, guardrails and provenance make every decision auditable.

MEASURED RESULT

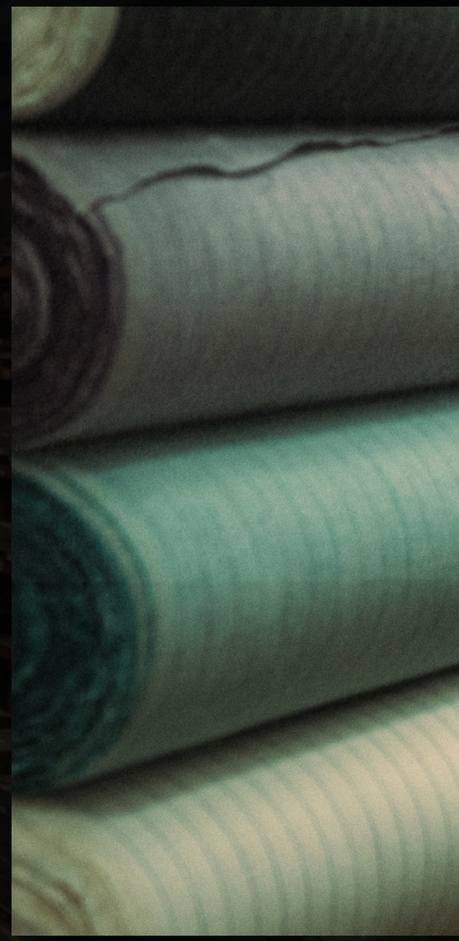
100% completion, tool, grounding, structured-output, constraint and routing rates across the deterministic evaluation suite.

Scope: synthetic demonstration data; Cloud Run SQLite is revision-local.



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References

[1] Google, Agent Development Kit documentation, 2026.
[2] CrewAI, Documentation, 2026.
[3] Linux Foundation, Agent2Agent Protocol Specification v0.3.
[4] Model Context Protocol, Specification, 2025–2026.
[6] Google Cloud, Cloud Run documentation.
[5] Pydantic, Documentation v2.
[7] FastAPI documentation.
[8] SAP IBP product overview.
[9] Kinaxis Maestro platform overview.
[10] o9 Solutions Digital Brain overview.

Repository:
github.com/thakuraastha/Bottleneck